

PAPER • OPEN ACCESS

Global plasma modeling of a magnetized high-frequency plasma source in low-pressure nitrogen and oxygen for air-breathing electric propulsion applications

To cite this article: Kryštof Mrózek *et al* 2021 *Plasma Sources Sci. Technol.* **30** 125007

View the [article online](#) for updates and enhancements.

You may also like

- [Electron dynamics in radio frequency magnetron sputtering argon discharges with a dielectric target](#)
Bocong Zheng, Yangyang Fu, Kelian Wang et al.
- [Electron scattering cross sections from \$NH_3\$: a comprehensive study based on \$R\$ -matrix method](#)
Yingqi Chen, Xianwu Jiang, Lufeng Yao et al.
- [Electron trapping efficiency of a magnetron sputtering cathode](#)
Mostafa Salahshoor

Global plasma modeling of a magnetized high-frequency plasma source in low-pressure nitrogen and oxygen for air-breathing electric propulsion applications

Kryštof Mrózek¹ , Tomáš Dytrych², Pavel Moliš³, Vladimír Dániel³ and Adam Obrusník^{1,2,4,*} 

¹ Department of Physical Electronics, Faculty of Science, Masaryk University, Brno, Czech Republic

² SpaceLabEU, Zlatníky, Czech Republic

³ VZLU a.s., Prague, Czech Republic

⁴ PlasmaSolve s.r.o., Brno, Czech Republic

E-mail: obrusnik@plasmasolve.com

Received 11 December 2020, revised 4 October 2021

Accepted for publication 4 November 2021

Published 20 December 2021



Abstract

This work presents a global plasma model of a gridded air-breathing electric propulsion concept based on magnetized high-frequency plasma operating in the pressure range of 10^{-3} Pa to 1 Pa. We illustrate that the global plasma model reproduces the experimental measurements of the extracted current over two orders of magnitude in pressure.

Consequently, we use the model to investigate the theoretical scalability of the plasma source, finding that the plasma source performance scales reasonably well with the average absorbed power per molecule, even though this scaling factor has its limits. The global model presented in this work is a model of a specific laboratory device and, in future, it can be adapted to very low Earth orbit conditions by adjusting the boundary conditions. The model was implemented using PlasmaSolve p3s-globalmodel software and the configuration file containing all the equations is provided to the community as supplementary material.

Keywords: plasma modeling, very low earth orbit, ECR plasma, air breathing electric propulsion, thruster

 Supplementary material for this article is available [online](#)

(Some figures may appear in colour only in the online journal)

* Author to whom any correspondence should be addressed.



Original content from this work may be used under the terms of the [Creative Commons Attribution 4.0 licence](#). Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

1. Introduction

Operating satellites in very low Earth orbit (VLEO)⁵ can yield several advantages. By reducing the mean altitude of a spacecraft, one can improve the accuracy of measurements of Earth monitoring devices such as, e.g. Earth observation satellites, weather satellites and spacecraft magnetometers. For example, reducing the operational altitude can double the resolution of an optical telescope, without improving parameters of the telescope itself. Operating a spacecraft closer to the surface of the Earth also improves the performance of telecommunication devices. Thus, orbiting satellites at VLEO have the potential to reduce the operational cost of a spacecraft while maintaining its performance or to increase its performance while maintaining the cost [1].

On the other hand, operating devices in VLEO poses unique challenges. The aerodynamic drag caused by the residual atmosphere becomes non-negligible below approximately 500 km of altitude. Therefore, any object in VLEO has the tendency to naturally de-orbit. The orbital lifetime of a spacecraft at around 400 km of altitude is typically a few years, but at altitudes below 300 km, the spacecraft will de-orbit in a matter of days or weeks [2]. In general, drag forces acting on a spacecraft must be compensated to prolong the lifetime of VLEO missions, thereby making them scientifically or commercially viable.

A promising and elegant technology concept for drag compensation in VLEO is air-breathing electric propulsion (ABEP). This technology is currently in the state of laboratory prototypes and concepts [3–5]. Since it uses the same residual atmosphere, which causes drag, as a propellant, there is no need for an internal supply of the propellant. An ABEP-based propulsion system combined with a reliable way of generating electrical energy on the orbit (e.g. solar panels) could potentially be able to extend the lifetime of missions in VLEO indefinitely⁶.

Pressure in VLEO varies approximately between 10^{-2} Pa at 100 km to 10^{-7} Pa at 400 km (see figure 1). Most ABEP concepts use plasma as a source of ions, which are accelerated to generate thrust [6], and most plasma sources are not able to operate at these pressures. Therefore, gas entering the thruster has to be compressed beforehand to enable plasma ignition and operation. This compression then decreases the amount of propellant available for propulsion and restricts maximum achievable thrust [8].

The plasma source studied in this work operates in a sort of ‘gray area’ between what is typically understood as electron-cyclotron resonance (ECR) plasma sources and magnetized radio-frequency (RF) sources. Firstly, we are using an electric field frequency of 435 MHz, while ECR is typically operated in the GHz region. At this frequency, the electromagnetic wave has a wavelength of approximately 69 cm. This is longer

than the characteristic dimensions of our reactor (7.5 cm), which would be more typical of a capacitively coupled plasma source. However, the plasma source in question does show resonance behavior with respect to the magnetic field. This is evidenced in the supplementary data (<https://stacks.iop.org/PSST/30/125007/mmedia>) to this paper, section 1, where the experimental measurements of current, produced by plasma, are shown as a function of the magnetic field. In this plot we do not see a sharp peak at magnetic field intensity corresponding to the ECR frequency, which would be typical of an ECR plasma source. However, we can still observe some resonance behavior—plasma is extinguished for a magnetic field below 13 mT and above 40 mT. For a magnetized capacitively Coupled Plasma (CCP) discharge, on the other hand, we would expect that the discharge would be sustained for any magnetic field above a certain threshold. Therefore, we have decided to call our plasma source a ‘magnetized high-frequency plasma source’, to prevent confusion. However, the physical processes inside the plasma should be similar to conventional ECR discharge.

Global models of non-magnetized RF discharges were described by Grondein [9] in iodine and Chabert [10] in xenon. Global Model (GM) of magnetized RF discharge was described by Lucken [11] for both iodine and xenon. While our plasma source is not ECR, many sources concerning ECR plasma sources are still relevant for us, since we are also dealing with a high-frequency electric field combined with a magnetic field. ECR plasma sources have been studied and modeled by the plasma physics community for several different applications. Kato [12] studied the charge-state distribution in low-pressure ECR plasma experimentally and compared it with theoretical predictions. Galatà [13] used numerical simulation to study the efficiency of ECR charge breeders. Mascali [14] employed a 3D particle-in-cell (PIC) method to study the influence of frequency and power absorption in ECR ion beam sources. More recently ion beam sources have been modeled by Mironov [15]. Hagelaar [16] developed 2D fluid simulation of low-pressure magnetized plasma. More relevant to our work is the global model of ECR plasma in a complex gas mixture used for plasma-enhanced chemical vapor deposition and developed by Yanguas-Gil [17]. Also relevant is a PIC model of an ECR thruster developed by Sanchez-Villar [18]. In particular, the PIC models used in the latter work are very advanced and provide a detailed insight into the interaction of the electromagnetic wave with the plasma. However, they all describe plasmas ignited in rare atomic gases, which is understandable considering the computational complexity of PIC. The added value in this work is the inclusion of a complex kinetic scheme of nitrogen and oxygen, which is immediately relevant for ABEP investigation. Furthermore, Lucken [19] studied the effect of a magnetic field on plasma confinement.

Several other concepts of ion sources have been proposed for ABEP application. Cifali [20] performed tests on an ABEP Hall thruster (HT), and Boeuf [21] presented a comprehensive review of HT and its modeling. Diamant [22] proposed the concept of two-stage ABEP based on HT, where the atoms are ionized in the first stage using ECR plasma and are accelerated in the second stage using the Hall effect. Hall ABEP

⁵ There is no formal definition of VLEO yet. In this paper we refer to Earth orbits where air resistance is not negligible—approximately below 450 km of altitude [1].

⁶ Realistically, the lifetime will be limited, e.g. by material degradation due to plasma exposure and VLEO atomic oxygen exposure.

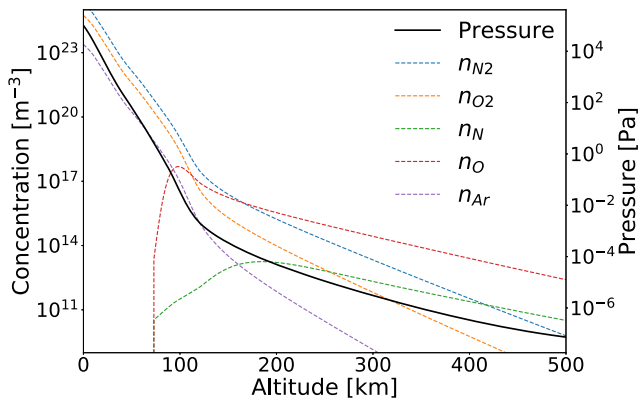


Figure 1. Number densities of dominant species (left axis) and total air pressure (right axis) as a function of altitude. Data taken from CCMC's (Community Coordinated Modeling Center) MSISE (Mass Spectrometer and Incoherent Scatter radar Extended) atmosphere model [7].

thrusters are also being tested by SITAEL [3] and JAXA [4]. More recently Romano [23] studied ABEP with a helicon-based inductive plasma thruster. A more exotic concept is that developed by IRS [24]. This concept uses inertial electrostatic confinement inside two spherical gridded electrodes to achieve relatively high plasma densities.

In this paper, we describe the implementation and validation of a global plasma model of an ABEP concept based on a magnetized high-frequency plasma source. The global plasma model describes a laboratory breadboard model capable of igniting plasma in mixtures of N_2 and O_2 . The numerical model shows good agreement with the experimental measurements of current extracted from the plasma, as discussed in section 5.1. Both the experimental data and the numerical model show that the plasma can be ignited at VLEO-relevant pressures of 10^{-3} Pa (assuming gas compression). The experimental validation of the global plasma model is followed by an extensive parametric study in section 5.5, which explores the theoretical scalability of the investigated ABEP concept. It is shown that, over a broad range of plasma volumes and absorbed plasma powers, the current density extracted from the plasma source scales with P/N , i.e. the average power absorbed by a molecule. Finally, we also investigate the effect of gas composition on the plasma source performance, which is presented in section 5.6. We also present the results of simulation of magnetic field and ECR surfaces, which allows us to better understand the physical phenomena inside our plasma source (section 2 in supplementary_figures.pdf).

The central aim of this work is to prepare a solid theoretical framework for studying the viability of in-orbit operation of a gridded ABEP system based on our high-frequency plasma source. The configuration file of the global model which contains all the physics is provided as supplementary data to this publication so that it can be adopted and extended by other researchers.

2. Experimental setup

Figure 2 shows a schematic picture of the thruster concept which is modeled and simulated in this work. The plasma is

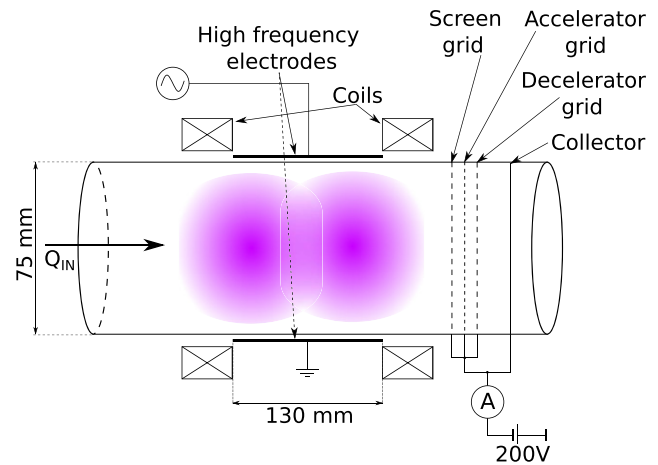


Figure 2. A schematic picture of the laboratory model of the ABEP thruster. The main part is a glass tube with gas inflow on one side and a series of grids (screen, accelerator, decelerator grid) and collector on the other. Two coils around the tube provide the magnetic field and electrodes on the outside of the tube supply an alternating electric field in the discharge. An ammeter is connected to the series of grids and the collector to measure the current produced by the discharge. After the ammeter there is a biased DC source.

ignited in a glass tube with the radius R and length L . An alternating electric field is supplied to the plasma by two planar electrodes placed on the outside of the tube. On the outside, there are also two coils generating a magnetic field parallel with the glass tube axis. One side of the tube represents an inlet through which gas is fed into the plasma, and the other side represents an outlet, through which gas can leave the system. On the latter, there are three grids: a screen grid, which has the same potential as plasma and extracts ions from plasma; an accelerator grid, which is set on a substantially lower potential and accelerates the ions out from the thruster; and a decelerator grid, which should prevent backstream of ions into the thruster [25]. Behind the grids there is a metal plate called a collector, which captures all ions leaving the thruster.

Synthetic air is supplied to the device through a mass flow controller and the plasma volume is evacuated to the desired pressures in the range of 10^{-3} Pa by a combination of membrane and turbomolecular pumps.

Diagnostics on the device are currently limited to current measurements. Specifically, we can measure the combined current flowing on all the grids and the collector. Only ions flowing on the collector would produce thrust in a real thruster, but we chose to measure the current this way because it is more convenient from an experimental perspective. After the ammeter there is another DC source. This component is here to keep the grids at the same potential as the walls. Without the compensation the walls would be on a floating potential and the voltage between them and grounded grids could influence the ions. The biasing voltage is about 200 V, which was enough to operate the ion-saturation region. The power generated by the power supply can also be measured (it is typically 4 W), although the fraction of the power which is actually absorbed by the plasma is not known.

Table 1. Parameters used in the model and their meanings.

L	13 cm	Length of glass tube with plasma
R	3.75 cm	Radius of glass tube with plasma
Q_{IN}	2.5 sccm $1.12 \cdot 10^{18} \text{ s}^{-1}$	Inflow of gas into the chamber
f_{N_2}	0.78	Fraction of molecular nitrogen in supplied gas mixture
f_{O_2}	0.22	Fraction of molecular oxygen in supplied gas mixture
P_{in}	0.3 W	Power input—fitting parameter
B	20 mT	Magnetic flux density
T_{amb}	310 K	Temperature of gas surrounding the plasma chamber
β_{n}	0.2	Transparency of screen grid for neutrals
β_{i}	0.6	Transparency of screen grid for ions
σ_{i}	$7.5 \times 10^{-19} \text{ m}^2$	Cross section for ion–neutral collisions

3. Model description

In this section, we discuss the mathematical formulation of the global model as well as the key assumptions that were made. The key parameters of the model together with their meanings and default values that were taken are summarized in table 1.

It should be highlighted that the model description in this section may be, as usual, subject to minor typos with major impact on the results. Therefore, we kindly refer any researcher attempting to reproduce our results to the model configuration file available as supplementary data. This model configuration file includes all the equations and physics as it was executed in machine-readable and human-readable format.

3.1. Plasma kinetic system and Boltzmann equation

For the global model, we have used the kinetic system of the N_2/O_2 mixture as described in Obrusník *et al* [26], which is based on [27, 28]. This kinetic system was originally used for modeling air plasma at 10^5 Pa and 300 K , but since the scheme was not developed for any specific type of plasma discharge and it contains large numbers of reactions we assume it is general enough to be used in our case.

By using this scheme we assume the gas fed into the thruster is composed only from nitrogen and oxygen. This is true in laboratory conditions, because the experimenter can select the composition of the gas used. It is not true for real conditions in orbit with the residual atmosphere containing additional species like argon but, as seen in figure 1, even at 100 km of altitude the concentration of argon is two orders of magnitude lower than the concentration of molecular oxygen. This difference only grows larger at higher altitudes; therefore, it is reasonable to assume that, by neglecting the influence of argon and other species, we only make small errors in our kinetic model.

We have made some updates to the kinetic system originally presented in [26]. Specifically, we assume that all effective temperatures of nitrogen species (variables T_{ionN} , T_{ionN_2} , T_{ionN_3} , T_{ionN_4} , T_{effN} , T_{effN_2} , T_{effN_3} and T_{effN_4} , see appendix A of [26]) are equal to neutral gas temperature. This assumption is motivated by the fact that the effective temperatures were originally used to develop a model of streamer discharges, but they are not relevant in the case of low temperature and low pressure plasma, such as magnetized high-frequency plasma described in this paper.

Table 2. Species included in the Boltzmann solver together with references for the cross-sectional data (taken from lxcat.net [30–32]).

Species	Numbers of reactions in [26]	Cross section reference
N_2	P001–P011, P023, P028	[33]
O_2	P012–P017, P025, P051, P056	[33]
N	P021	[34]
O	P019, P020, P022	[34]
NO	P027, P029, P030, P052	[35]
N_2O	P031	[36]
O_3	P053, P054	[34]
$\text{O}_2(\text{a}^1)$	P018, P026, P055	[34]
$\text{N}_2(\text{A}^3)$	P024	[34]
N_2^+	P050	[37, 38]

Since our plasma is heavily magnetized, we cannot assume Maxwell–Boltzmann distribution for electron energies. The shape of the electron energy distribution function (EEDF) can have a great impact on rate coefficients of electron-impact reactions (ionization, excitation, dissociation, recombination). Therefore, we have decided to use the Boltzmann equation solver BOLSIG+ to calculate the EEDF [29]. Its knowledge enables us to more accurately describe rate coefficients of electron-impact reactions with known cross sections. It is important to note here that the two-term approximation BOLSIG+ uses to solve the Boltzmann equation might not be the best option to calculate the EEDF in a crossed magnetic and alternating electric field. For this reason we are looking for ways to calculate the EEDF using the more robust Monte-Carlo-based solver. However, at this point BOLSIG+ is still the best option for us.

The list of used cross section databases leveraged in this work can be found in table 2 and the BOLSIG+ configuration file is provided in the supplementary data of this paper (bolsig_configuration.txt). The EEDF calculated using BOLSIG+ is shown in figure 3(a) together with the corresponding Maxwellian EEDF. The steady-state electron temperature in our model is 4.5 eV , which corresponds to a mean electron energy of 6.75 eV . We have also calculated rate coefficients of electron impact ionizations of N_2 , N , O_2 and O for both Maxwellian EEDF and EEDF calculated using BOLSIG+. These rate coefficients are shown in figure 3(b).

We can see that our EEDF deviates quite dramatically from Maxwellian shape, especially for energies above 20 eV .

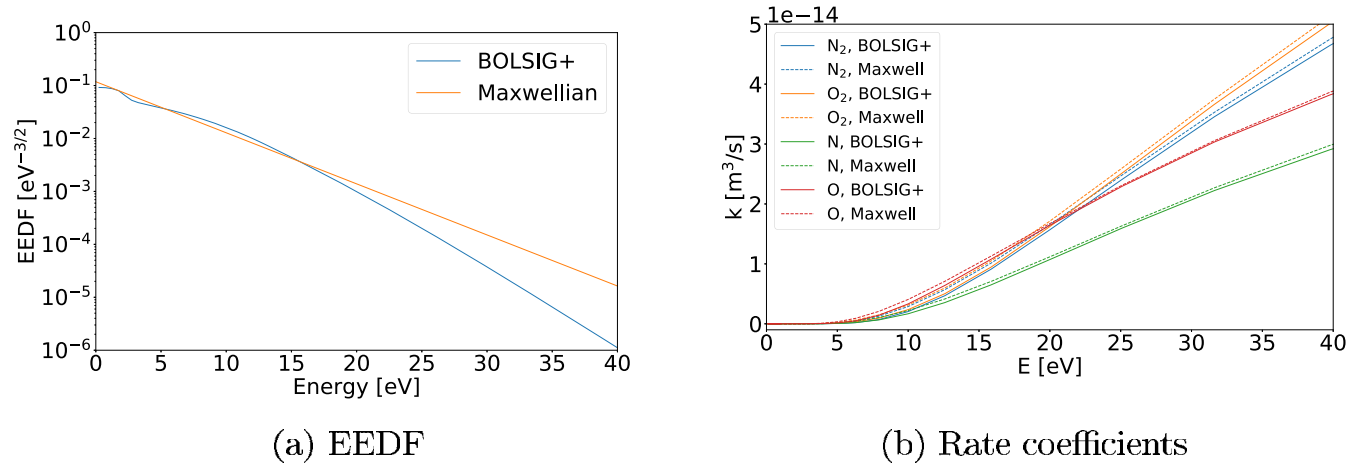


Figure 3. EEDF calculated using BOLSIG+ with energy 6.75 eV and the Maxwellian EEDF for the corresponding electron temperature 4.5 eV together with rate coefficients of electron impact ionization of selected species calculated from both EEDFs.

Table 3. Species (neutrals and ions) used in the reaction scheme.

Neutrals	N_2 , $N_2(A^3)$, $N_2(B^3)$, $N_2(a^1)$, $N_2(C^3)$, N , $N(^2D)$, $N(^2P)$, O_2 , $O_2(a^1)$, $O_2(b^1)$, $O_2(4.5 \text{ eV})$, O , $O(^1D)$, $O(^1S)$, O_3 , NO , N_2O , NO_2 , NO_3 , N_2O_5
Positive ions	N^+ , N_2^+ , N_3^+ , N_4^+ , O^+ , O_2^+ , O_4^+ , NO^+ , N_2O^+ , NO_2^+ , $N_2O_2^+$, $N_2^+(B^2)$
Negative ions	O^- , O_2^- , O_3^- , O_4^- , NO^- , N_2O^- , NO_2^- , NO_3^-

It is also worth noting that our EEDF have larger values between 5 and 15 eV. However, this change in EEDF from the Maxwellian shape does not have a great impact on actual rate coefficients. From this, we conclude that our model is not very sensitive to the change in EEDF.

The kinetic system contains 42 species—21 neutral species, 12 positive ions, 8 negative ions and electrons, see table 3. For each of these species, except electrons, a balance equation is solved in the form

$$\frac{\partial n_s}{\partial t} = S_{s,r} + S_{s,w} + S_{s,in} + S_{s,out}, \quad (1)$$

where s is the index of the species. The terms $S_{s,w}$, $S_{s,in}$ and $S_{s,out}$ are species-specific and denote wall loss/gain, inflow and outflow of species, respectively. The exact expressions for these terms are listed below.

The reaction source term $S_{s,r}$ is added to each equation based on the kinetic scheme. This term has the following form

$$S_{s,r} = \sum_{i \in RS} \chi_{i,s} k_i \prod_{j \in Re} n_j. \quad (2)$$

Here, RS denotes a set of all reactions considered for a given species and k_i denotes the rate coefficient for a given reaction. Also, $\chi_{i,s}$ is the number of particles of given species produced

(positive values) or lost (negative values) by the reaction. The index j runs over all reactants Re in a given reaction and n_j denotes the number density of a given reactant.

In addition to the volume processes, there are other important channels through which species can be introduced into the plasma or escape it. Three different types of species can be found in the plasma—neutrals, positive ions and negative ions—each of them is treated differently.

3.1.1. Charged particle balance. Negative ions are the simplest to describe. Due to the fact that plasma has a higher electric potential than the walls and grids surrounding it, we assume that the negative ions are confined to plasma and cannot leave through the plasma sheath. Therefore, only terms appearing on the right hand sides of the differential equations, describing the population of negative ions, are expressed in equation (2).

Positive ions can, in contrast, leave through the sheath, and can either leave the system completely or can be neutralized on the walls and grid and return to the plasma as neutral species. Therefore, we must introduce corresponding terms into the differential equations.

To be able to correctly describe the flux of ions leaving the plasma, we need to know the concentration and velocity of particles at the edge of the plasma sheath. Since we are using a volume-averaged global model we have no information about spatial distribution of ions. To estimate the change in ion concentration from an averaged value to the value at the edge, we employ so-called h factors [40]. These factors provide us with a ratio of ion density in the center of the discharge and on its edge. Due to the presence of oxygen gas in the plasma, we need to take the electronegativity of the discharge into account [42, 43]. We only take into consideration the two-region low-pressure regime of electronegative discharge. Then we can define the factors in the longitudinal h_L and radial h_R direction

$$h_L = \frac{0.86}{(1 + \alpha_0) \sqrt{3 + \frac{\eta L}{2\lambda_i}}}, \quad (3)$$

$$h_R = \frac{0.8f_b}{(1 + \alpha_0)\sqrt{4 + \frac{\eta R}{\lambda_i}}}. \quad (4)$$

Here, $\alpha_0 = n_-/n_e$ denotes the electronegativity of plasma, $\lambda_i = 1/(n_g\sigma_i)$ signifies the mean free path of ions in neutral gas, and $\eta = 2T_+/(T_+ + T_-)$ is used as a simplifying nomenclature [44]. We use n_- to denote the concentration of all negative ions, n_e for the concentration of electrons, n_g for the concentration of neutrals, T_- for the temperature of negative ions, T_+ for the temperature of positive ions (we assume both these temperatures to be equal to the temperature of neutral gas T_{gas} —see section 3.2) and σ_i for the ion–neutral cross section.

Furthermore, the coefficient f_b in equation (4) is defined as $f_b = (1 + \omega_{c,i}\tau)^{-1}$, where $\omega_{c,i} = qB/m_i$ is the average cyclotron frequency of ions and $\tau = \lambda_i/v_{th}$ is the mean free time of ions between collisions with neutrals. The symbol v_{th} denotes the mean thermal velocity of ions (see equation (19)). We are using the average ion mass m and their thermal velocity v_{th} , weighted by the concentration of a given ion, to calculate cyclotron frequency or mean free time. The coefficient f_b describes the confinement of ions in a magnetic field. We only apply this coefficient in the direction perpendicular to the magnetic field (this is the radial direction in our case) [45].

Using these factors the flux of ions, leaving the plasma in the longitudinal or radial direction, respectively, can be written as follows [17]

$$\Gamma_{i,L} = v_{b,i}n_i h_L, \quad (5)$$

$$\Gamma_{i,R} = v_{b,i}n_i h_R. \quad (6)$$

Here, n_i denotes the volume-averaged concentration of a given ion and $v_{b,i}$ denotes its Bohm velocity. The Bohm velocity in electronegative plasma can be written as [40]

$$v_{b,i} = \sqrt{\frac{qT_e(1 + \alpha_0)}{M_i(1 + \alpha_0 r_-)}}, \quad (7)$$

where $r_- = T_e/T_-$ and T_e is the electron temperature. Both temperatures are expressed in eV. Here, it is important to note that using Bohm velocity is not consistent with section 3.1. Bohm velocity is derived assuming Maxwellian EEDF. That is not our case, as we can see in figure 3(a). However, at low energies (up to 20 eV) the shape of our EEDF is very close to the Maxwellian shape; thus we consider the approximation of ion velocities with Bohm velocity reasonable. Also, at the same time, due to the nature of volume-averaged global models, we have no better way to describe the ion velocity.

Due to the specific implementation of this model it is more convenient to define effective areas through which the ions can leave the plasma and include h factors in them (so we do not define fluxes using h factors). Thus, we have defined areas

$$A_{\text{eff},i,e} = \pi R^2 h_L \beta_i, \quad (8)$$

$$A_{\text{eff},i,w} = 2\pi RL h_R + \pi R^2 h_L (1 - \beta_i). \quad (9)$$

The area $A_{\text{eff},i,e}$ denotes the transparent area of the grid through which ions can leave the plasma and the area $A_{\text{eff},i,w}$ denotes the area upon which ions can be neutralized (either the wall of

the thruster or the acceleration grid). The symbol β_i signifies the transparency of the grid for ions (in general, transparency can be different for ions and neutrals). This means the ratio of ions which pass the grid upon leaving the plasma through the exhaust. Currently, we can either estimate the transparency of the grid or possibly use it as a fitting parameter of the model. To get more reliable values, direct simulation Monte Carlo (DSMC) simulation of particle flow upon the grid would need to be employed.

Using equations (8) and (9) we can write the term describing the loss of any given ion leaving plasma through the grid as follows

$$S_{i,e} = -n_i v_{b,i} \frac{A_{\text{eff},i,e}}{V_{\text{pl}}}, \quad (10)$$

where $V_{\text{pl}} = \pi R^2 L$ denotes the volume of plasma. In a similar fashion we can write the terms for ions hitting the insides of the thruster and consequently neutralizing into neutral species (assuming every ion hitting the walls gets neutralized)

$$S_{i,w} = -n_i v_{b,i} \frac{A_{\text{eff},i,w}}{V_{\text{pl}}}. \quad (11)$$

All things considered, the right-hand side of all differential equations, describing the populations of positive ions, comprises expressions (2), (10) and (11).

It is important to note here that while expressions (10) and (11) act like a loss term for differential equations of ions, expression (11) also acts like a source term for specific neutral species because ions that are neutralized at the wall return to the plasma as neutrals. In table A1 there is a list of all considered reactions of ions on walls. Keep in mind that conservation laws have to be satisfied while implementing these reactions. This means that, for example in the reaction N004, the term for N_2 generation has to be twice as large as the term for N_4^+ loss.

Finally, the electron density in the model n_e is calculated from the plasma quasineutrality assumption

$$n_e = n_+ - n_-, \quad (12)$$

where n_+ and n_- denote the concentration of all positive and negative ions, respectively.

3.1.2. Neutral particle balance. In the laboratory breadboard model, a steady flow rate Q_{IN} (in s^{-1}) of N_2/O_2 mixture is supplied into the discharge chamber. We can write the source term for inflow of a given gas as follows

$$S_{n,\text{in}} = \frac{Q_{\text{IN}} f_n}{V_{\text{pl}}}. \quad (13)$$

Here, f_n denotes the fraction of given neutral species in the supplied gas mixture. In our case we are supplying only N_2 and O_2 , and so $f_{N_2} + f_{O_2} = 1$.

At the same time the neutral gas inside the thruster is pumped out with pumping speed S_{pump} (in $\text{m}^3 \text{s}^{-1}$). We assume all neutral species are perfectly mixed inside the chamber and, therefore, the rate with which a specific species is leaving the

chamber is a function of the total pumping speed and the number density. For a given species we can express the pumping loss term as

$$S_{n,\text{out}} = -\frac{S_{\text{pump}}n_n}{V_{\text{pl}}}. \quad (14)$$

Neutral species are also introduced into plasma through volume reactions (equation (2)) and neutralization of ions on the chamber walls (table A1 and equation (11)), and can also undergo recombination and deexcitation on the walls.

Recombination and deexcitation reactions have neutral species as reactants and, when they hit a wall, it will react with a certain probability. This sets the neutral species apart from the ions which, in our model, always recombine upon hitting the wall. We define the total effective area $A_{\text{eff},n,e}$ through which neutrals can leave the plasma and the effective area $A_{\text{eff},n,w}$ upon which neutrals can recombine or undergo deexcitation $A_{\text{eff},n,w}$ as

$$A_{\text{eff},n,e} = \pi R^2 \beta_n, \quad (15)$$

$$A_{\text{eff},n,w} = 2\pi RL + \pi R^2(1 - \beta_n), \quad (16)$$

where β_n is transparency of the grid for neutrals.

Now we can write the term for loss or gain of neutral species through wall reactions as

$$S_{n,w} = \Gamma_{n,r} \frac{A_{\text{eff},n,w}}{V_{\text{pl}}}, \quad (17)$$

where $\Gamma_{n,r}$ is the flux of neutral particles which undergo recombination or deexcitation (we assume that the concentration of neutrals is constant throughout the plasma—there is no h factor for neutrals) [42]

$$\Gamma_{n,r} = \frac{\gamma}{2(2 - \gamma)} n_n v_{\text{th}}, \quad (18)$$

where γ is either the recombination coefficient (γ_{rec}), defined in table A2, or the deexcitation coefficient (γ_{deex}), defined in table A3. Furthermore, we use the concentration of the given species n_n and its thermal velocity

$$v_{\text{th},n} = \sqrt{\frac{8k_B T_n}{\pi m_n}}, \quad (19)$$

where $T_n (= T_{\text{gas}})$ is the temperature of a given species and m_n is its mass.

Similar to equation (11) for ions, equation (17) describes a loss term for one neutral species and a gain term for another one according to tables A2 and A3. It is important to add this term to each differential equation with the correct sign and also the correct number of particles lost/gained.

In total, the right-hand side of a neutral balance equation includes the contributions of the following processes:

- a source term for volume reactions—equation (2),
- inflow and outflow of neutral species from plasma—equations (13) and (14),
- neutralization of ions on the walls—equation (11),
- recombination and deexcitation on the walls—equation (17).

3.2. Energy balance

Energy is fed into the plasma through an alternating electric field operating in the RF range. This way, the energy is first transferred only to electrons. From here it can dissipate from the system through several channels. Part of the energy is transferred to populations of neutral species, heating them. Part of the energy is used for excitation and ionization processes. Part is lost when ions leave the plasma, and the rest is turned into thermal energy of the electron population. We are solving the following equation for the electron energy density (whereby we can calculate electron temperature)

$$\frac{\partial E_e}{\partial t} = \frac{\partial}{\partial t} \left(\frac{3}{2} T_e n_e \right) = S_{E,e,\text{abs}} + S_{E,e,\text{el}} + S_{E,e,\text{in}} + S_{E,e,\text{b}}. \quad (20)$$

Power deposition to the plasma is expressed as

$$S_{E,e,\text{abs}} = \frac{P_{\text{in}}}{qV_{\text{pl}}}, \quad (21)$$

where the power absorbed by the plasma P_{in} is used as a fitting parameter in our model and is assumed to be constant over a broad range of pressures. Although this is generally a strong assumption, comparison of the model results with experimental measurements confirms it. By doing this we also assume that the power absorption in plasma does not change during temporal evolution of the discharge. It is reasonable to assume that this parameter depends on the applied magnetic field, so we cannot assume the power absorption stays constant with the changing magnetic field (see section 5.2).

Transfer of energy between the electron population and populations of neutral species happens primarily through elastic collisions. We can express the source term for transfer of energy between the electrons and populations of neutral species as [40]

$$S_{E,e,\text{el}} = -3m_e k_B (T_e - T_{\text{gas}}) n_e \sum_n \frac{n_n k_{\text{el},n}}{m_n}. \quad (22)$$

Here, the sum is over all considered neutral species, $k_{\text{el},n}$ is the rate coefficient of elastic collisions between electrons and a given neutral species, n_n denotes the concentration of that neutral and m_n is its mass. We are using values of $k_{\text{el},n}$ calculated using BOLSIG+. To compute this term we consider only populations of N_2 , N , O_2 and O , because concentrations of other neutral species are usually an order of magnitude lower than these four species⁷. Also, T_{gas} denotes the temperature of neutral gas. We assume that all neutral species (and also ions) have the same temperature.

Electrons also lose energy through inelastic collisions with neutral particles (like ionization and excitation). The loss of energy through a given reaction can be written as

$$S_{E,e,\text{in}} = -\sum_r k_r E_{\text{th},r} \prod_{j \in \text{RS}} n_j. \quad (23)$$

⁷ N and O have low concentrations in our laboratory conditions, but they are prevalent in orbit, so we decided to include them here to make the model general; see figures 5 and 6 in section 5.3.

We include all electron-induced ionization and excitation reactions in the sum, including rotational and vibrational excitations of molecular gases, which are likely dominant energy loss channels. The symbol k_r is again the rate coefficient of a given reaction, E_{th} is the threshold energy of the reaction and n_j denotes the concentration of a reactant in a given reaction. Naturally, k_r strongly depends on the electron temperature and it is calculated using BOLSIG+.

Furthermore, the electron population as a whole loses energy when an ion leaves the plasma. The amount of electron energy $E_{b,i}$ effectively lost per each ion leaving the plasma through a boundary can be expressed as [39, 40]

$$E_{b,i} = 2T_e + \varphi + \frac{T_e}{2} \ln \left(\frac{m_i}{2\pi m_e} \right) \approx \frac{7}{2} T_e. \quad (24)$$

Here, the first term is the energy lost when an electron leaves the plasma (each time an ion leaves, an electron must also leave to maintain quasineutrality of the plasma), the second term $\varphi = T_e/2$ [40] is the kinetic energy of an ion at the sheath boundary and the last term is energy gained by an ion while traversing the sheath. From this energy we can express the source term for the electron energy loss caused by ions leaving the plasma as

$$S_{E,e,b} = - \sum_i \frac{(A_{eff,i,e} + A_{eff,i,w})}{V_{pl}} u_{b,i} n_i E_{b,i}. \quad (25)$$

We sum over all positive ions present in the plasma.

Besides the equation for the temperature of electrons the model also solves the equation for the energy density E_{gas} of neutral gas

$$\begin{aligned} \frac{\partial E_{gas}}{\partial t} = \frac{\partial}{\partial t} \left(\frac{3}{2} k_B T_{gas} n_t \right) = -S_{E,e,el} + S_{E,n,r} \\ + S_{E,n,w} + S_{E,n,a}, \end{aligned} \quad (26)$$

where n_t is the sum of concentrations of all neutral particles and ions—we are including ions due to the assumption of thermal equilibrium between all heavy particles in plasma. So T_{gas} is actually the temperature of all neutral species and ions combined, but in this paper we refer to it as a neutral gas temperature, because neutral species have much higher number densities than ions. This means we assume the ion temperature is equal to the temperature of neutral gas. We have made this assumption because it is difficult to estimate actual ion temperature. In reality this assumption is an important source of uncertainty in our model. We have carried out uncertainty analysis, so we are able to at least estimate the effect of this uncertainty (see section 5.2).

Heavy particles gain energy through elastic collisions with electrons, as described by equation (22). They can also gain or lose energy through volume reactions, depending on the enthalpy of reactants and products. We can describe this source term, similarly to the term (23), as

$$S_{E,n,r} = \sum_r k_r H_r \prod_{j \in RS} n_j. \quad (27)$$

Here, we sum over all reactions, in which electrons are not involved, and H_r denotes the enthalpy difference of reactants and products of a given reaction.

Neutral particles and positive ions can also hit a wall, during which they are assumed to thermalize with the wall and neutralize (in the case of ions). This leads to effective cooling of heavy particles inside plasma (ions return as neutrals) and heating of the wall. In this model we assume that the discharge chamber wall is immediately cooled by the surrounding gas and kept at constant temperature. This assumption is valid in a laboratory model of the thruster, but might not be valid for conditions in orbit, due to low pressure (therefore low heat conductivity) of the surrounding atmosphere.

We can then define the total flux of neutral particles $\Gamma_{t,n}$ and ions $\Gamma_{t,i}$ onto the wall using the mean thermal velocity of neutrals and the Bohm velocity of ions

$$\Gamma_{t,n} = \sum_n n_n v_{th,n}, \quad (28)$$

$$\Gamma_{t,i} = \sum_i n_i v_{b,i}. \quad (29)$$

Here, the summation runs over all neutral or all positive species, respectively. Using these fluxes we can express the source term for the loss of energy from heavy particle populations due to wall interactions as

$$\begin{aligned} S_{E,n,w} = - \frac{3}{2V_{pl}} k_B (T_{gas} - T_{wall}) \\ \times (\Gamma_{t,n} A_{eff,n,w} + \Gamma_{t,i} A_{eff,i,w}), \end{aligned} \quad (30)$$

where T_{wall} is the temperature of the chamber walls, which is equal to the temperature of the surrounding gas $T_{amb} = 300$ K (i.e. room temperature in our model).

Finally, the temperature of heavy species is decreased by cool gas flowing into the chamber and warm gas being pumped out—so-called advective loss. With gas inflow Q_{IN} into the chamber (in s^{-1}), we know the number of particles flowing into the chamber per second. Knowing this, we can write the source term for energy loss due to inflow of gas as

$$S_{E,n,a} = - \frac{3}{2V_{pl}} k_B (T_{gas} - T_{wall}) Q_{IN}. \quad (31)$$

4. Model implementation

The model is implemented using proprietary software called the p3s-globalmodel developed by PlasmaSolve. All physics of the global model are included in a single configuration file and the software compiles the fully symbolic definitions of a non-linear system of ordinary differential equations into optimized functions with fast evaluation. To solve the system of equations, we use the BDF implicit solver or CVODA solver. The BDF solver has a superior performance at relatively high relative tolerances of $>10^{-7}$, while CVODA tends to be faster at lower relative tolerances $\leq 10^{-7}$.

Overall, the calculation of the model with 43 independent variables and 700 reactions to steady state (typically 100 s of

physical time) takes about 60 s of computational time. This makes the global plasma model a very good tool for rapid prototyping of the plasma source and for carrying out extensive parametric studies.

The configuration file of the simulation is provided in the supplementary data (simulation_configuration.txt).

5. Results and discussion

5.1. Model validation

To confirm that our plasma global model actually reflects the physics happening in the ABEP concept, we validated the model with experimental results. We compare the measured extracted current and current calculated using the global model as a function of pressure. Our choice of this quantity is motivated by two reasons: it can be measured directly and without too many assumptions, and is also closely related to the thrust of a hypothetical ABEP system.

Current measured on grids is influenced by their transparency β_1 (see equation (9)). Since we do not know the value of this parameter, trying to compare experimentally measured current on grids with current calculated by the model is difficult. Instead we are measuring the current on all the grids and the collector. This makes the use of current measurement to validate the model more reliable. Also, we can place the collector into different positions behind the grids. This enables us to estimate loss of current on the thruster walls, but at this time we are not able to include this loss in the model.

The total ion current on all the grids and the collector can be calculated from the global model using the Bohm velocity of ions and their concentration on the plasma boundary (see section 3.1.1) using

$$I = \sum_i n_i v_{b,i} A_{\text{eff,curr}}, \quad (32)$$

where we sum over all positive species and $A_{\text{eff,curr}} = \pi R^2 h_L$ is an effective area for calculating the measured current.

At the current state of the global model, we are not able to describe power losses between the power supply and the plasma itself. For this reason P_{in} represents the power absorbed by the plasma, not the actual experimental power supply, and we are using it only as a fitting parameter of the model. At the moment a new breadboard model is being built which enables us to measure power losses and also enables us to measure current generated by plasma more reliably. As a result of this, we hope we will be able to better describe the power input in our model more accurately.

With this new breadboard model we also intend to carry out experimental studies on the influence of magnetic field on the efficiency of power transport into the plasma and on magnetic confinement. We want to combine these experimental results with parametric studies performed using the global model and hopefully gain more insight into the effect of magnetic field on the discharge.

During the measurement the electric field with a frequency of 435 MHz was supplied to the electrodes. The strength of magnetic flux density in the center of the glass tube and

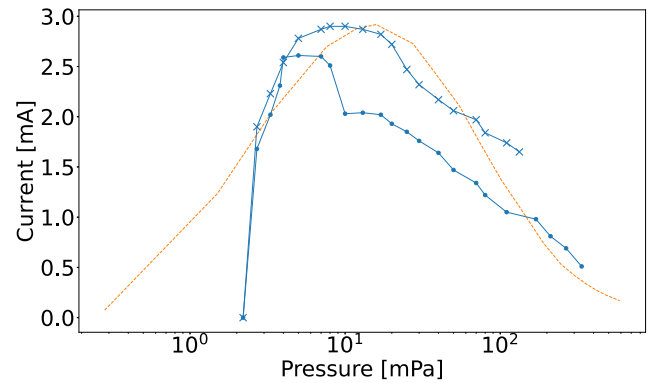


Figure 4. Experimental validation for 435 Mhz and 22 mT with impedance adjustment. The two blue curves correspond to two independent measurements and they give an idea of the uncertainty of the experimental data.

coil was 22 mT. Note that this value differs slightly from the value of the magnetic field we used in the model—20 mT. This is because we measured the magnetic field in the center of one coil. The magnetic field in the region of the ECR surface will be slightly lower (see figure 2(b) in supplementary_figures.pdf).

Some inflow of gases was set and after the plasma reached steady state, the pressure and current on all the grids and the collector were measured. Between each measurement, the electrical circuit was adjusted, until the maximum current was obtained at the given pressure. This justifies the assumption of keeping P_{in} from equation (21) constant and using it as a fitting parameter.

The data series was measured for two different distances of the collector from the grids. The experimental results together with the simulation results are shown in figure 4. We can see that over a range of pressures between 2 and 200 mPa, the difference between currents measured for different collector positions is about 1 mA and is also roughly constant. This implies that the ions are mainly due to the divergence of the plasma beam behind the grids and ions getting neutralized on the walls of the thruster.

Simulation was implemented using the parameters given in table 1 and fitted using the input power $P_{\text{in}} = 0.3$ W. Figure 4 illustrates good agreement of the model and experiment over roughly two orders of magnitude in pressure. This is an important result which qualifies the model for further simulation-aided engineering of the ABEP concept.

An important difference between the simulated and experimental results is the position of the maximum of the current. This could be caused by the fact that the pressure was measured near the inlet and outlet and then averaged. The actual pressure in the plasma is probably not the average between these two values because the grids and the current collector act as a considerable vacuum resistance and the pressure profile between the inlet and outlet is likely not linear.

There is large deviation between the experimental results and results from our global model in the low-pressure region (below 2 mPa). It is possible that assumption of constant power coupling efficacy breaks down in this region, but this topic

needs to be investigated further, using more advanced power absorption measurements. We still choose to show the results of our model even outside the validated region (2–200 mPa), so the trends in our model can be better observed.

The simulated curve also has a slightly different slope in the high-pressure region, although this is within the experimental uncertainty and it will also be the subject of further investigations.

5.2. Main sources of uncertainty

The following section identifies the main uncertainties in the presented global model. The global modeling approach brings an inherent uncertainty caused by the fact that the model is volume averaged and does not consider spatial distribution of the plasma or the electromagnetic fields. Other uncertainty arises from the fact that input data (reaction rates, cross sections) are not available.

A major uncertainty source arising from the volume averaging was outlined in section 3.2 and it is the power absorbed by the plasma P_{in} . At this point, we have no way to calculate this quantity independently and it has to be treated as a fitting parameter with the assumption it remains constant for different pressures. However, P_{in} does not remain constant for a varying magnetic field. This can be evidenced in figure 2 in the supplementary_figures.pdf, which shows that the shape and position of the ‘ECR surface’ changes qualitatively with the applied magnetic field. This will certainly impact the power absorption efficiency. In other words, for a different value of magnetic field intensity, P_{in} would have to be fitted independently.

Another source of uncertainty is the fact that the global model is not state-resolved. Vibrational state can have a great effect on many molecular processes. For example, according to Napari *et al* [41], rate coefficients of dissociative electron attachment on O_2 can differ by up to a factor 30, depending on the vibrational state of O_2 . However, making a state-resolved global plasma model is a tremendous task, which might currently be impossible due to the lack of rate constants for all the vibrational states of O_2 and N_2 .

Lastly, we should consider the uncertainty arising from our assumptions about the ion temperature. We assume that ions have the same temperature as neutral gas as the volume-averaged global model does not give us any better way of estimating it. To assess the sensitivity of the model with respect to this assumption, we refer the reader to figure 3 in the supplementary_figures.pdf. We conclude that the impact is rather minor but should be considered in future research.

5.3. Number densities of neutrals and ions

In a plasma, an abundance of species might be produced. It is therefore useful to have at least an estimate of the plasma composition. We have performed parametric studies with varying inflow of gas into the chamber and looked at the concentrations of the most important species in the plasma at different pressures. Figure 5 shows the number densities of four most important species— N_2 , N, O_2 and O. Furthermore, figure 6 shows the densities of positive ions of each of these species and three negative oxygen ions. These results were obtained

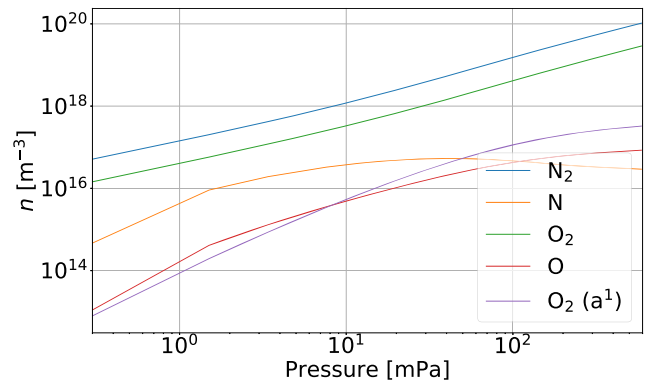


Figure 5. Number densities of selected neutral species as a function of pressure.

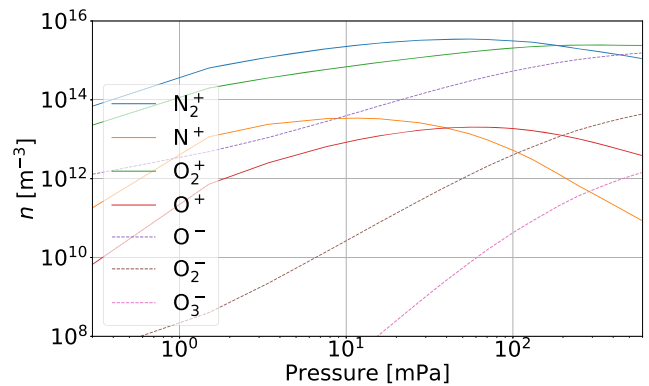


Figure 6. Number densities of selected ions as a function of pressure.

from simulation with the power input $P_{in} = 0.3$ W, plasma radius $R = 3$ cm, plasma length $L = 10$ cm and magnetic field $B = 20$ mT. These parameters apply for all simulation results presented in this paper, unless otherwise specified.

As expected, the concentration of molecular nitrogen and oxygen increases with the pressure. This is simply the consequence of greater inflow of these gases into the plasma. On the other hand, the number densities of atomic oxygen and nitrogen are not monotonous and we see that the maximum density of N is at around 30 mPa. We do not see the maximum of O in our range but it is reasonable to assume that there is one. This behavior probably occurs due to larger recombination rates (three body collisions) at higher pressures.

Similar behavior can be seen for all charged species. Negative ions reach the maximum number density at pressures at least an order of magnitude higher than positive ions. It is worth noting that O^- reaches density higher than the N_2^+ ion at pressure over 500 mPa and might even become the dominant ion species for even higher pressures.

5.4. Discharge ignition

One of the key parameters of the thruster is a range of conditions under which it can operate. With regard to a hypothetical ABEP at VLEO, we are primarily interested in pressure limits of plasma operation at a relatively low available power.

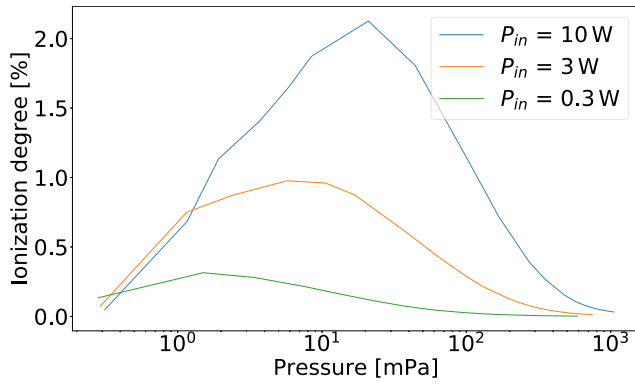


Figure 7. Simulated dependency of the ionization degree in the discharge on pressure. Default plasma source dimensions were used.

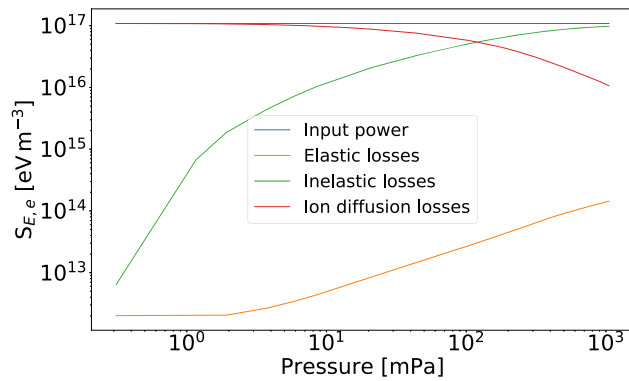


Figure 8. Energy gain and losses from electron populations through different channels (i.e. different terms in equation (20)) as a function of pressure.

Figure 7 shows the ionization degree as a function of the pressure for three values of absorbed plasma power. The ionization degree ID is calculated as a fraction of all positive ions n_+ in the plasma

$$ID = \frac{n_+}{n_g + n_+ + n_-}, \quad (33)$$

where n_g is the concentration of all neutral species.

We can see that for a given power input there is always an optimal pressure, at which the ionization degree reaches a maximum. If we increase the pressure further, the ionization degree starts to decrease. This is due to more energy from the electron population being lost by inelastic collisions, as evidenced in figure 8. Here, we can see that, at low pressures, almost all energy is lost through ion diffusion from plasma (see the term $S_{E,e,b}$ in equations (20) and (25)). This is because at these pressures there is a small number of electrons and they have a high temperature (figure 9). This means every electron (drawn out by ions) which leaves the plasma results in relatively high energy losses.

With increasing pressure energy loss through this channel is decreasing and losses due to inelastic collision are becoming more important. Eventually, at around 1 Pa, inelastic collisions become the dominant energy-loss channel and they are the reason plasma decays at high pressure. Energy loss due to elastic collision also increases, but is still far less important.

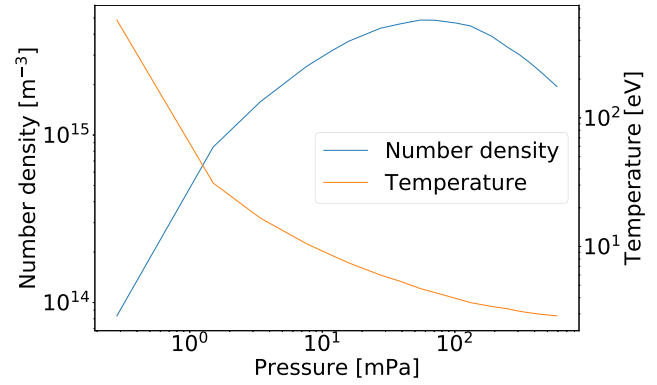


Figure 9. Electron number density and temperature as a function of pressure. We define temperature as $T_e = \frac{2}{3}\bar{E}$, where $\bar{E}$ is mean electron energy in eV.

At lower pressures we observe a much steeper decrease in the electron density and ultimately plasma extinction. Even if electrons are confined very well in this case (low electron–neutral collision frequency), they do not have sufficient density and subsequently ionization frequency and the discharge is not self-sustained. In figure 9 we observe a steep decrease in electron density for low pressures (we see a decrease for high pressure for reasons described in the previous paragraph).

We can also see that the temperature increases with decreasing pressure. This is caused by the fact that we are still supplying the same amount of energy into the discharge, but it has to be absorbed by a smaller number of electrons. For pressures under 1 mPa it even gets higher than 100 eV. This most likely is not a real effect, but is caused instead by the break down of our description of the power absorption into the plasma. Power coupling efficiency would probably be smaller for plasmas with density this low, meaning our assumption of constant P_{in} is no longer valid in this range.

Because the pressure at 100 km of altitude is of the order of 10^{-2} Pa it might be feasible to operate ABEP at this altitude, given sufficient power input. However, pressure decreases quite rapidly with altitude (10^{-7} Pa at 400 km). Therefore, to operate the ABEP thruster presented in this paper, at higher altitudes, some sort of compression between the outside and inside of the thruster would certainly need to be employed.

In the follow-up work, we intend to carry out a detailed analysis of the drag and thrust at various altitudes using the presented ABEP concept.

5.5. Plasma source scaling and its limits

For future optimization, it is useful to know to what extent is it possible to scale the plasma source to a larger volume and power and, therefore, thrust. The global model is a very suitable tool for exploring such a multi-dimensional parametric space.

To this end a series of parametric studies was performed, simultaneously varying three variables—the power input P_{in} , radius R of the plasma and its length L . Frequency and magnetic field strength were not varied since they would likely

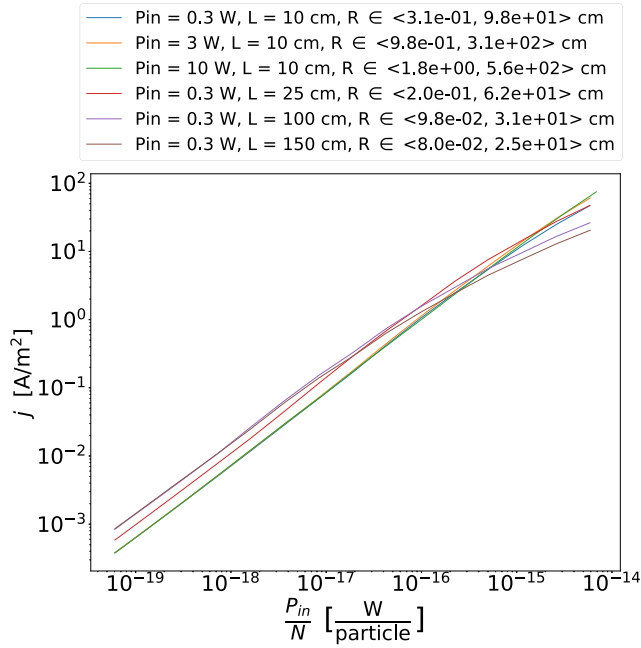


Figure 10. Current density as a function of power input per heavy particle in plasma for several combinations of P_{in} and L with varying R .

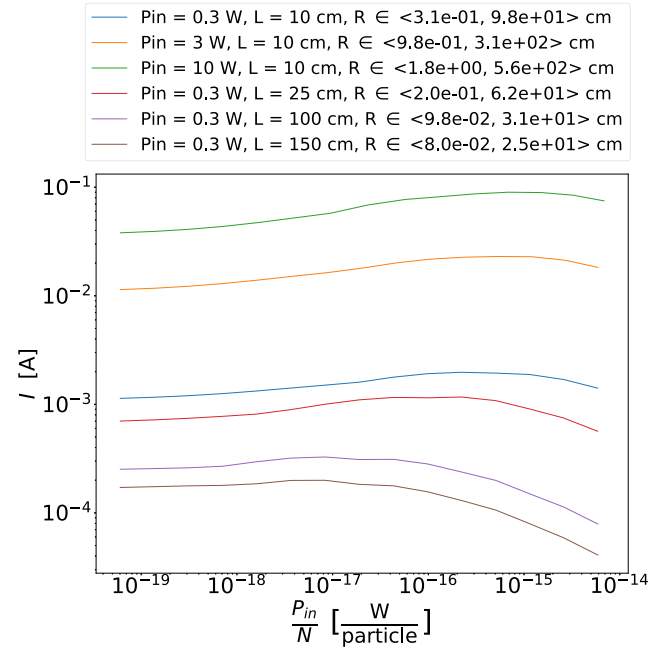


Figure 11. Current as a function of power input per heavy particle in plasma for several combinations of P_{in} and L with varying R .

influence the power coupling efficiency, which is assumed constant in our model. We have varied these parameters over a broad range of values, more than two orders of magnitude. In some cases we used values which are clearly not physical; for example, a thruster radius lower than 1 mm or larger than 5 m. While we realize it is not possible to build the thruster of these sizes, we still believe it useful to show the results of these studies, so the trends in our model and limits of its scaling are more clearly visible.

The figures below always show the current density at the extraction grids j and the total current I that can be extracted from the discharge. The current is calculated using formula (32). It is linked with current density simply as

$$I = A_{\text{eff, curr}} j. \quad (34)$$

Both quantities are plotted as a function of the power per particle, P/N , defined using the total number density of all species (except electrons) n_{tot} as

$$\frac{P}{N} = \frac{P_{in}}{V_p n_{\text{tot}}}. \quad (35)$$

Firstly, let us investigate the case when R was varied while P_{in} and L were kept constant. Looking at figure 10, we see that the current density scales very well with P/N . The simulations predict that at low P/N , a longer tube increases the current density while, at high powers, the model that has shorter tubes will be favorable, which is a surprising conclusion. The reason is that for a large P/N and L , the plasma radius R is of the order of millimeters and the radial plasma confinement no longer prevents the radial wall loss of charged species.

However, looking at the current plot in figure 11, we clearly see that a large plasma radius and shorter length are

favorable over long plasma with small radius. This is only logical because the grid area grows with R^2 and so does the total current. The conclusion that the current density extracted by the grids scales well with P/N is important for future feasibility assessment of the gridded ABEP concept with a magnetized high-frequency plasma source. Interestingly, the current curves in figure 11, especially those for $L = 10$ cm configurations, are relatively flat. This means that if the plasma radius is increased, the decreasing ion density is well compensated by the increasing extraction area.

In the next study, we maintained R and L constant while varying P_{in} . In figure 12, we again see a good current scaling of the current density with P/N . Again, we see that the current density scaling breaks at higher P/N , where the current density for large-diameter plasmas is smaller. This can be attributed to a different discharge regime because a large P/N and large R requires plasma power in the kilowatt range. It is likely that in this high-power range, different volume phenomena become influential, e.g. positive–negative ion recombination.

The situation is different for the total extracted current, as seen in figure 13. For most of the investigated region, a large-diameter plasma source seems favorable over small-diameter thrusters and this difference vanishes only at high P/N .

The overall outcome of the parametric studies can be summarized by saying that upscaling and downscaling of thrusters, based on technology described in this paper, shows great promise with the power per heavy particle P/N being the main scaling parameter for the current density. The total extracted current can consequently be controlled by changing the radius of the plasma source, with larger radii naturally leading to higher currents. Proper tradeoff analysis between theoretical thrust and drag is envisioned in follow-up work.

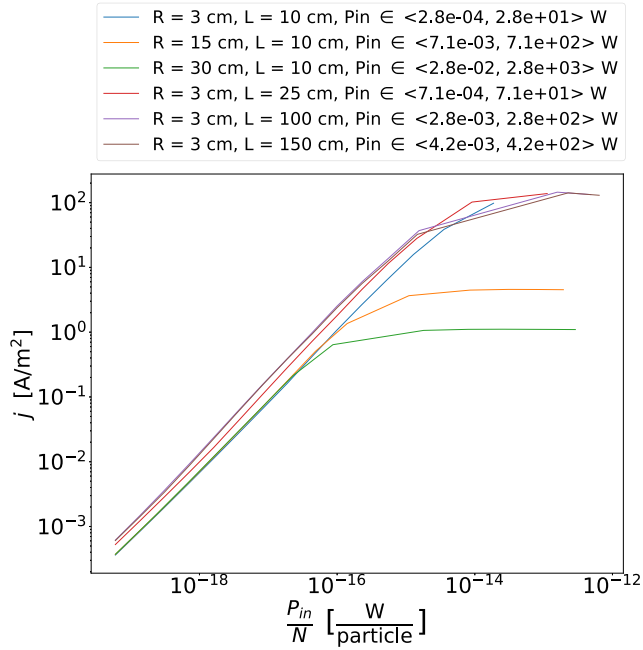


Figure 12. Current density as a function of power input per heavy particle in plasma for several combinations of R and L with varying P_{in} .

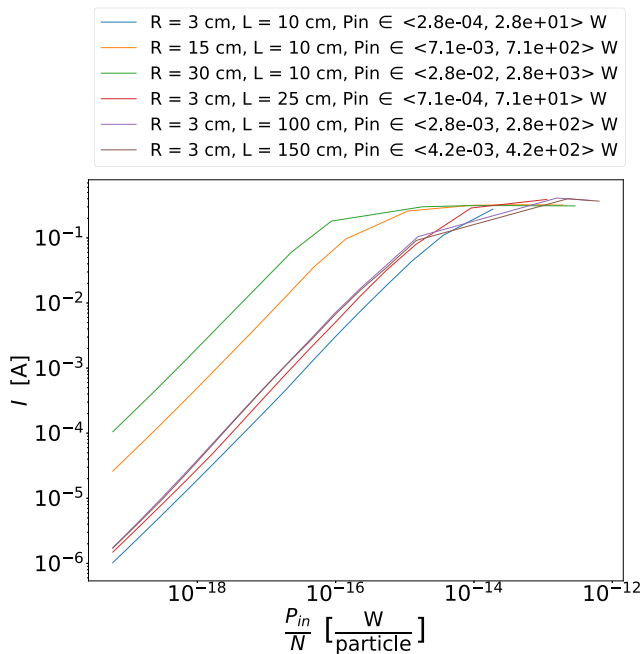


Figure 13. Current as a function of power input per heavy particle in plasma for several combinations of R and L with varying P_{in} .

5.6. Influence of gas composition

Apart from the total pressure, the gas composition also changes at VLEO. As illustrated in figure 1, the number density of N_2 (O_2) changes from approximately 10^{19} m^{-3} (10^{18} m^{-3}) at 100 km of altitude to 10^{10} m^{-3} (10^8 m^{-3}) at 500 km. Also, below 100 km atomic nitrogen and oxygen are practically nonexistent, but it starts to increase and atomic oxygen

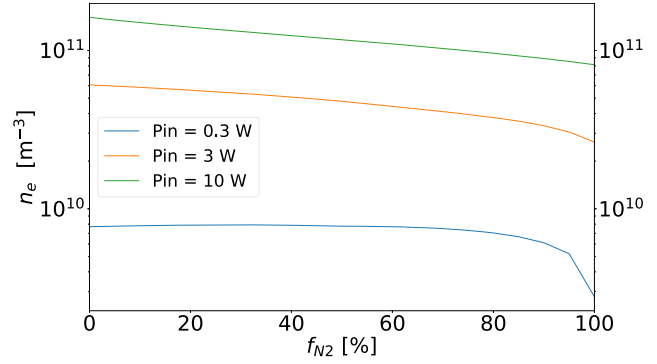


Figure 14. Electron number density as a function of N_2 content in supplied gas for different values of power input.

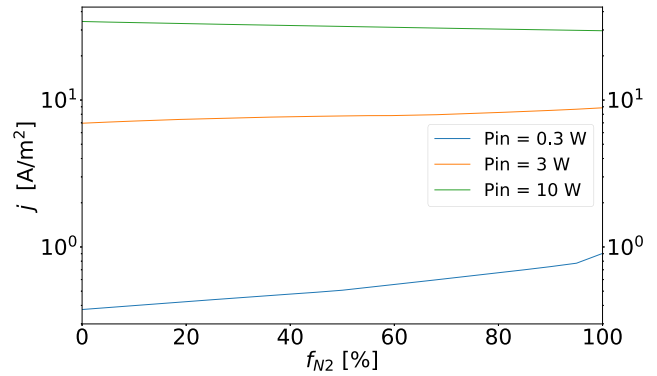


Figure 15. Current density as a function of N_2 content in supplied gas for different values of power input.

becomes more prevalent than molecular nitrogen at around 180 km and atomic nitrogen becomes more common than molecular nitrogen at 420 km.

In this work, we performed parametric studies where we varied the content of molecular nitrogen in supplied gas between 0 and 100% for three different plasma powers. The corresponding change in the electron density is shown in figure 14 and the change in current density is shown in figure 15.

As illustrated in figure 14 the electron number density decreases with N_2 content. This is due to lower ionization energy of both molecular and atomic oxygen (compared to nitrogen). More interesting is the fact that current density (figure 15) can either increase or decrease based on power input. This is caused by a combination of two effects. In more nitrogen-rich plasma, there are fewer positive ions, therefore lower current. But at the same time there is also less oxygen and thus fewer negative ions, which lowers the confinement of positively charged particles (described by the h_L parameter—equation (3)). So, in nitrogen plasma, there are fewer ions, but these ions are more likely to reach the plasma boundary. Which effect dominates depends on the power input. For higher power the oxygen plasma is less electronegative to begin with, so a decrease in negative ions is not sufficient to compensate for a decrease in positive ions.

All of the claims in the previous paragraph are supported by figure 16, where the electronegativity is plotted as a function

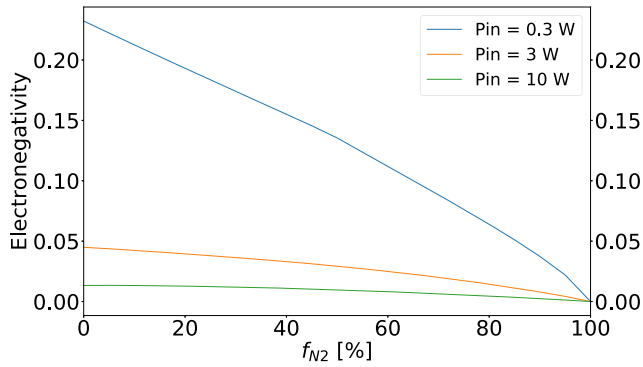


Figure 16. Electronegativity of a discharge as a function of N_2 content in supplied gas for different values of power input.

of the nitrogen supply. Obviously, we can observe a decrease in electronegativity with decreasing oxygen content and we can also see that there are large differences between different powers. For 10 W of input power, the discharge has almost zero electronegativity, even for pure oxygen plasma, and the electronegativity increases with decreasing power.

6. Conclusion

We presented a global plasma model describing a magnetized high-frequency plasma source operating at low pressures of 10^{-3} to 10^0 Pa. This plasma source is intended as a source of ionization in an ABEP concept.

Firstly, we have illustrated that the global model shows good agreement with experimentally measured positive ion current collected from the plasma. This good agreement is reached over the range of two orders of magnitude in gas pressure. The additional information deduced from the model is the ionic composition of the plasma over the two orders of magnitude in pressure, where it was discovered that molecular ions dominate in the laboratory breadboard model, which will certainly be different at VLEO, and that the concentration of negative ions only becomes non-negligible at the higher pressures, towards 1 Pa.

Secondly, both the numerical model and the experiments consistently suggest that igniting the plasma at pressures lower than 10^{-3} Pa is not feasible. Even though the radial plasma confinement becomes very strong at the low pressures, the ionization rate is not high enough to ensure self-sustained plasma. Therefore, we conclude that a magnetized high-frequency plasma source with reasonable dimensions is only applicable to ABEP systems if the internal pressure in the thruster will exceed approximately 5 mPa.

Next, the theoretical limits of the scaling of the plasma source were investigated in an extensive parametric study. It was discovered that, over a broad range of conditions, the current density j scales with the average power absorbed per

Table A1. A list of reactions of ions with thruster walls producing neutrals. Gain or loss terms (or rate coefficients of the reactions) from differential equations can be calculated using equation (11).

	Reaction
N001	Wall + N^+ → wall + N
N002	Wall + N_2^+ → wall + N_2
N003	Wall + N_3^+ → wall + N + N_2
N004	Wall + N_4^+ → wall + N_2 + N_2
N005	Wall + O^+ → wall + O
N006	Wall + O_2^+ → wall + O_2
N007	Wall + O_4^+ → wall + O_2 + O_2
N008	Wall + NO^+ → wall + NO
N009	Wall + N_2O^+ → wall + N_2O
N010	Wall + NO_2^+ → wall + NO_2
N011	Wall + $N_2O_2^+$ → wall + N_2 + O_2
N012	Wall + $N_2^+(B^2)$ → wall + N_2

heavy particle P/N . Limits of this scaling were identified for plasma sources with a large radius (15–30 cm) operating in the kilowatt power range. Finally, we looked at the total extracted current I , concluding that larger plasma source diameters are beneficial, especially in the low- P/N region.

In follow-up work, we intend to complement the modeling with more detailed experimental characterization. In terms of modeling, the global plasma model of laboratory magnetized high-frequency discharge will be extrapolated to VLEO conditions by changing the gas composition. The pressure inside the plasma source will be obtained from 3D DSMC simulation of gas compression, from which we will also be able to deduce drag at various altitudes. With these tools, we will be able to complete a simulation-driven feasibility study for the presented magnetized high-frequency ABEP concept.

Acknowledgment

This research was supported by the project LM2018097 funded by the Ministry of Education, Youth and Sports of the Czech Republic. The authors also gratefully acknowledge the financial support of CQK Holding group represented by Michal Pajr who funded the simulation work and VZLU a.s. for the decision to internally fund the experimental work.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Appendix A. Tables of reactions on the walls

(See tables A1–A3)

Table A2. A list of neutral recombination reactions on thruster walls together with used recombination coefficients and their references. We assume that the coefficient for $O(^1S)$ is equal to one for $O(^1D)$.

	Reaction	γ_{rec}	Reference
R001	Wall + N + N $\rightarrow$ wall + N ₂	0.07	[46]
R002	Wall + N(² D) + N(² D) $\rightarrow$ wall + N ₂	0.07	[46]
R003	Wall + N(² P) + N(² P) $\rightarrow$ wall + N ₂	0.07	[46]
R004	Wall + O + O $\rightarrow$ wall + O ₂	0.17	[47]
R005	Wall + O(¹ D) + O(¹ D) $\rightarrow$ wall + O ₂	0.17	[47]
R006	Wall + O(¹ S) + O(¹ S) $\rightarrow$ wall + O ₂	0.17	[47]

Table A3. A list of neutral deexcitation reactions on thruster walls together with used deexcitation coefficients and their references. We assume the coefficient for all N₂ species to be the same as for N₂(A³), for all O₂ to be the same as for O₂(4.5 eV) and for O(¹S) to be the same as for O(¹D).

	Reaction	γ_{deex}	Reference
D001	Wall + N ₂ (A ³) $\rightarrow$ wall + N ₂	1.00	[46]
D002	Wall + N ₂ (a ¹) $\rightarrow$ wall + N ₂	1.00	[46]
D003	Wall + N ₂ (B ³) $\rightarrow$ wall + N ₂	1.00	[46]
D004	Wall + N ₂ (C ³) $\rightarrow$ wall + N ₂	1.00	[46]
D005	Wall + O ₂ (a ¹) $\rightarrow$ wall + O ₂	0.007	[47]
D006	Wall + O ₂ (b ¹) $\rightarrow$ wall + O ₂	0.007	[47]
D007	Wall + O ₂ (4.5 eV) $\rightarrow$ wall + O ₂	0.07	[46]
D008	Wall + N(² D) $\rightarrow$ wall + N	0.93	[46]
D009	Wall + N(² P) $\rightarrow$ wall + N	0.93	[46]
D010	Wall + O(¹ D) $\rightarrow$ wall + O	0.17	[47]
D011	Wall + O(¹ S) $\rightarrow$ wall + O	0.17	[47]

ORCID iDs

Kryštof Mrózek  <https://orcid.org/0000-0002-9520-7906>

Adam Obrusník  <https://orcid.org/0000-0002-8415-4909>

References

- [1] Virgili-Llop J, Roberts P, Hao Z, Ramio L and Beauplet V 2014 Very low Earth orbit mission concepts for Earth observation. Benefits and challenges *12th Reinventing Space Conf. 2014* (Belfast) 1–18
- [2] Schönherr T, Komurasaki K, Romano F, Massuti-Ballester B and Herdrich G 2015 *IEEE Trans. Plasma Sci.* **43** 287–94
- [3] Andreussi T, Ferrato E, Giannetti V, Piragino A, Cifali G and Andrenucci M 2019 Development status and way forward of SITEAL's air-breathing electric propulsion *AIAA Propulsion and Energy Forum 2019* (Denver)
- [4] Romano F, Binder T, Herdrich G and Fasoulas S 2016 *Intake Design for an Atmosphere-Breathing Electric Propulsion System, Space Propulsion 2016* (Rome)
- [5] Cara D Di, del Amo G, Santovincenzo A, Dominiquez B, Arcioni M, Caldwell A and Roma I J 2007 RAM electric propulsion for low Earth orbit operation: an ESA study *30th Int. Electric Propulsion 2007* (Florence)
- [6] Romano F, Binder T, Boxberger A, Smith K and Roberts P 2017 *Literature Review of ABEP Systems D 4.1 Discoverer*
- [7] Hedin A E 1987 *J. Geophys. Res.* **92** 4649–62
- [8] Barral S, Cifali G, Albertoni R, Andrenucci M and Walpot L 2015 Conceptual design of an air-breathing electric propulsion system *IEPC-2015-271/ISTS-2015-B-271* (Hyogo-Kobe)
- [9] Grondein P, Lafleur T, Chabert P and Aanesland A 2016 *Phys. Plasmas* **23** 033514
- [10] Chabert P, Arancibia Monreal J, Bredin J, Popelier L and Aanesland A 2012 *Phys. Plasmas* **19** 073512
- [11] Lucken R, Marmuse F, Bourdon A, Chabert P and Tavant A 2019 Global model of a magnetized ion thruster with xenon and iodine *36th IEPC* (Vienna)
- [12] Kato Y, Matsumoto K and Ishii S 1992 *J. Phys. Soc. Japan* **62** 1221–32
- [13] Galatà A, Patti G, Celona L, Mascali D, Neri L and Torrisi G 2016 *Rev. Sci. Instrum.* **87** 02B505
- [14] Mascali D, Torrisi G, Neri L, Sorbello G, Castro G, Celona L and Gammio S 2015 *Eur. Phys. J. D* **69** 27
- [15] Mironov V, Bogomolov S, Bondarchenko A, Efremov A, Loginov V and Pugachev D 2020 *J. Inst.* **15** P10030
- [16] Hagelaar G J M 2007 *Plasma Sources Sci. Technol.* **16** S57–66
- [17] Yanguas-Gil A, Cotrino J and González-Elipe A R 2007 *J. Phys. D: Appl. Phys.* **40** 3411–22
- [18] Sanchez-Villar A, Zhou J, Merino M and Ahedo E 2019 PIC/fluid/wave simulation of the plasma discharge in an ECR plasma thruster *36th IEPC 2019* (Vienna)
- [19] Lucken R, Tavant A, Bourdon A, Lieberman M A and Chabert P 2020 *Plasma Sources Sci. Technol.* **29** 065014
- [20] Cifali G, Misuri T, Rossetti P, Andrenucci D, Feili D and Lotz B 2011 Experimental characterization of HET and RIT with atmospheric propellants *32nd IEPC 2011* (Wiesbaden)
- [21] Boeuf J-P 2017 *J. Appl. Phys.* **121** 011101
- [22] Diamant K D 2009 Microwave cathode for air breathing electric propulsion *31st IEPC 2009* (Michigan)
- [23] Romano F et al 2020 *Acta Astronaut.* **176** 476–83
- [24] Chan Y A and Herdrich G 2018 Breakthrough of inertial electrostatic confinement concept for advanced space propulsion *69th IAC 2018* (Bremen)
- [25] Sangregorio M, Xie K, Wang N, Guo N and Zhang Z 2018 *Chin. J. Aeronaut.* **31** 1635–49
- [26] Obrusník A, Bilek P, Hoder T, Šimek M and Bonaventura Z 2018 *Plasma Sources Sci. Technol.* **27** 085013
- [27] Flitti A and Pancheshnyi S 2009 *Eur. Phys. J. Appl. Phys.* **45** 21001
- [28] Capitelli M, Ferreira C M, Gordiets B F and Osipov A I 2000 *Plasma Kinetics in Atmospheric Gases* (Berlin: Springer)
- [29] Hagelaar G J M and Pitchford L C 2005 *Plasma Sources Sci. Technol.* **14** 722–33
- [30] Pancheshnyi S, Biagi S, Bordage M, Hagelaar G, Morgan W, Phelps A and Pitchford L 2012 *Che-m. Phys.* **398** 148–53
- [31] Pitchford L C et al 2017 *Plasma Process. Polym.* **14** 1600098
- [32] Carbone E, Graef W, Hagelaar G, Boer D, Hopkins M M, Stephens J C, Yee B T, Pancheshnyi S, van Dijk J and Pitchford L 2021 *Atoms* **9** 16
- [33] Phelps database 2020 www.lxcat.net (retrieved 14 January 2020)
- [34] Morgan database 2020 www.lxcat.net (retrieved 14 January 2020)
- [35] Itikawa database 2020 www.lxcat.net (retrieved 14 January 2020)
- [36] Hayashi database 2020 www.lxcat.net (retrieved 14 January 2020)
- [37] eMol-LeHavre database 2020 www.lxcat.net (retrieved 14 January 2020)
- [38] Little D A, Chakrabarti K K, Mezei J Z and Schneider I F 2014 *Phys. Rev. A* **90** 052705
- [39] Bosi F J 2019 *Phys. Plasmas* **26** 033510

- [40] Lieberman M A and Lichtenberg A J 1994 *Principles of Plasma Discharges and Materials Processing* (New York: Wiley)
- [41] Napari M, Tarvainen O, Kinnunen S, Arstila K, Julin J, Fjellvåg Ø S, Weibye K, Nilsen O and Sajavaara T 2017 *J. Phys. D: Appl. Phys.* **50** 095201
- [42] Toneli D A, Pessoa R S, Roberto M and Gudmundsson J T 2015 *J. Phys. D: Appl. Phys.* **48** 325202
- [43] Kim S, Lieberman M A, Lichtenberg A J and Gudmundsson J T 2006 *J. Vac. Sci. Technol. A* **24** 2025
- [44] Monahan D D and Turner M M 2008 *Plasma Sources Sci. Technol.* **17** 045003
- [45] Cheetham A D and Rayner J P 1998 *J. Vac. Sci. Technol. A* **16** 2777
- [46] Thorsteinsson E G and Gudmundsson J T 2009 *Plasma Sources Sci. Technol.* **18** 045001
- [47] Gudmundsson J T, Kouznetsov I G, Patel K K and Lieberman M A 2001 *J. Phys. D: Appl. Phys.* **34** 1100–9